BANKING CLIENTS DATA ANALYTICS

REPORT

Workflow: Excel → SQL → Power BI

1. Overview

This project analyzes banking client data to understand customer segments, deposits, loans, business lending, advisor performance and risk. The analysis follows an end-to-end data analyst workflow: data preparation in Excel, analytical queries in SQL, and dashboard visualization in Power BI.

2. Excel – Data Preparation

The raw banking client dataset was prepared in Excel before analysis. The final dataset contains customer demographics, income, deposits, loans, credit-card balances, business lending, relationship type, loyalty classification, fee structure, risk weighting and advisor information.

Key preparation activities:

- Reviewed column names and data types.
- Checked customer, demographic and financial fields for consistency.
- Prepared the clean banking-client dataset for SQL analysis and Power BI.

3. SQL – Business Analysis

SQL views were created to answer the main business questions. The project includes segment-value analysis, risk-versus-loan/credit analysis and customer deposit ranking.

4. Power BI Dashboard

The Power BI dashboard converts the SQL outputs into interactive business visuals.

Implemented views include:

- KPI cards: Total Clients, Total Deposits, Total Loans and Total Credit Card Balance .
- Banking Relationship analysis using deposits, loans and business lending.
- Loyalty Classification × Fee Structure segment analysis.
- Risk Weighting analysis for average loans and credit-card balances.
- Top customers by net worth/deposits using ranking or Top N filtering.
- Investment Advisor analysis using client count and financial measures.
- Slicers for Nationality, Fee Structure, Gender, Loyalty Classification, Banking Relationship and Age to support interactive analysis.

5. Key Business Questions

1. What does the client base look like by age, gender and nationality?

- **Overall Dataset**
  - The dataset contains 2,940 clients across 27 columns after cleaning process.
  - There are 4 banking relationship types, 4 loyalty classifications and 3 categories of fee structure.
  - Risk weighting ranges from 1 to 5,useful for risk segmentation.
  - There are 22 investment advisors, giving enough scope for advisor-performance analysis.

Count of Client ID	Column Labels		
Nationality	Female	Male	Grand Total
African	86	86	172
American	231	268	499
Asian	368	369	737
Australian	120	130	250
European	669	613	1282
Grand Total	**1474**	**1466**	**2940**

Age Band	Count of Client ID
<30	551
30-49	868
50+	1521
Grand Total	**2940**

- **Clients overview**
  - There are 2,940 clients; **1,474** are females and **1,466** are males from various nationalities but top 2 nationalities are European(1282 clients) and Asian(737 clients).
  - The client base is strongly concentrated among mid-age and older customers.

2. Which loyalty classification and fee structure segments hold the most deposits?

	loyalty_classification	fee_structure	clients	avg_income	total_deposits
1	Jade	High	657	174180.684840183	477797215.65
2	Jade	Mid	417	165045.552014389	270669012.12
3	Silver	High	355	175047.493070423	240750608.63
4	Gold	High	294	165973.741836735	190813519.48
5	Jade	Low	235	179378.762808511	167522550.24
6	Silver	Mid	248	174382.318306452	156874186.37
7	Gold	Mid	169	179543.366982248	124427150.35
8	Silver	Low	149	178094.103825503	97263049.63
9	Platinum	High	137	166944.744890511	82355500.85
10	Gold	Low	108	152509.834444444	66394651.94
11	Platinum	Mid	114	161450.651842105	63269539.27
12	Platinum	Low	57	163524.802982456	43570105.63

Jade + High Fee is the strongest segment, with 657 clients and approximately $47.77 crore in total deposits and $50.44 crore in net worth.

Here are the top 3 :

Segment	Clients	Avg. Income	Total Deposits
Jade – High	657	$1.74 lakh	$47.77 crore
Jade – Mid	417	$1.65 lakh	$27.06 crore
Silver – High	355	$1.75 lakh	$24.07 crore

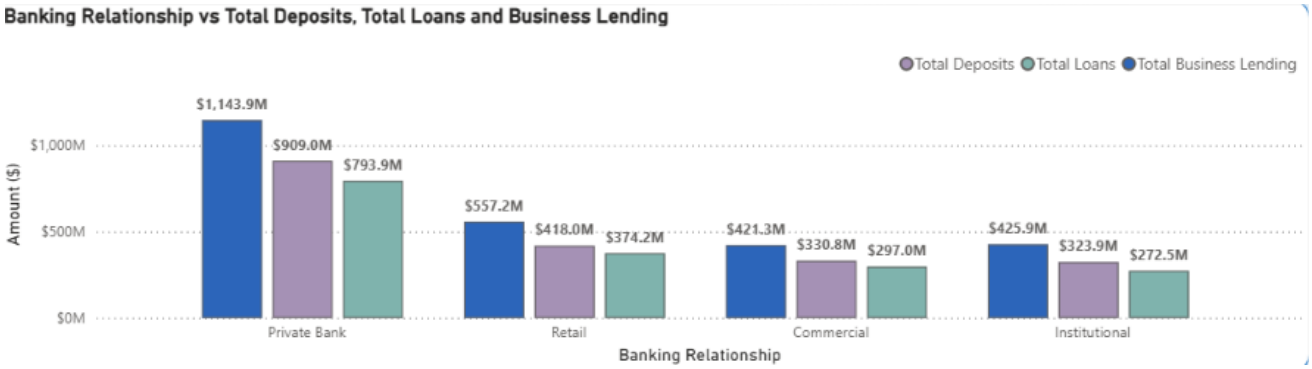
3. How does risk weighting relate to loans and credit-card balances?

There is a strong upward pattern: as Risk Weighting increases from 1 to 5, average bank loans and average credit-card balances also increase. Risk 5 has the highest average loan ($9.68 lakh) and credit-card balance ($5259), while Risk 1 has the lowest.

	Risk_Weighting	clients	avg_loans	avg_cc_balance	avg_income
1	1	820	333019.19	1876.44	77785.01
2	2	1197	576653.49	3082.46	159383.55
3	3	446	787324.86	4105.31	245078.55
4	4	319	846789.64	4511.82	284731.13
5	5	158	968534.66	5259.76	309159.66

4. Which banking relationship drives the most deposits, loans and business lending?

Private Bank is the leading relationship type, with 1,323 clients and approximately $90.90 crore in deposits. It also has the highest total business lending at approximately $79.39 crore and total loans of $114.3 crore.



Relationship	Clients	Total Deposits	Business Lending
Private Bank	1,323	$90.90 crore	$114.39 crore
Retail	648	$41.80 crore	$55.72 crore
Commercial	484	$33.08 crore	$42.13 crore
Institutional	485	$32.39 crore	$42.59 crore

5. Who are the top customers by net worth?

Top 10 Clients

Name	Total Net Worth
Daniel Gomez	$8.60M
Harry Burns	$5.99M
Gerald Harrison	$5.67M
John Frazier	$5.67M
Martin Marshall	$5.56M
Ronald Reed	$5.55M
Michael Bishop	$5.43M
Timothy Hunter	$5.26M
Frank Fox	$5.25M
Harry Gibson	$5.08M
Total	**$58.04M**

Net worth of top 10 clients varies from **$8.60M** to **$5.08M.**

Combine net worth of top 10 clients is $58.04M(2.85%) out of total net worth $2032.5M and Damiel Gomez is on the top with net worth of 8.60M.

6. Who are the top investment advisors by net worth managed?

Top 10 Advisors by Net Worth Managed

Investment Advisor	Net Worth
Peter Castillo	$137.83M
Lawrence Sanchez	$132.18M
Joe Carroll	$132.04M
Jeremy Porter	$129.91M
Victor Dean	$129.72M
Daniel Carroll	$127.81M
Victor Gutierrez	$122.27M
Eric Shaw	$113.05M
Carl Anderson	$112.69M
Nicholas Ward	$107.73M
Total	**$1,245.22M**

Top 10 investment advisors are managing worth around $1245.22M .

Peter Castillo is on the top with $137.83M.

6. Final Findings

- ✓ Private Bank is the most valuable relationship segment by total deposits and business lending.
- ✓ Jade–High Fee is the strongest loyalty/fee combination by total deposits.
- ✓ Risk Weighting shows a clear positive relationship with average bank loans and credit-card balances.
- ✓ Higher-risk customers should receive closer monitoring because of greater lending and credit exposure.
- ✓ Customer ranking and advisor-level analysis can help identify high-value customers and support relationship-management decisions.
- ✓ Power BI provides an interactive layer for management to explore these findings by customer and segment dimensions.

7. Conclusion

The project demonstrates a complete data analytics workflow from Excel data preparation to SQL-based business analysis and Power BI visualization. The analysis provides actionable insights into customer value, deposits, lending exposure, risk and relationship segments, supporting data-driven banking decisions.